FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 15-12-2019

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

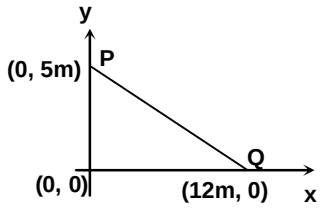
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A particle is executing simple harmonic motion along a line $12y + 5x = 60$ only in between the extreme points 'P' and 'Q'. At $t = 0$, the particle starts its motion from the extreme point 'P'. The time period of simple harmonic motion executed by the particle is 8 sec. Then choose the correct option(s).



(A) The position vector of the particle at time t is given by $\vec{r} = \left[12\sin^2\left(\frac{\pi t}{8}\right)\hat{i} + 5\cos^2\left(\frac{\pi t}{8}\right)\hat{j}\right]$ m.

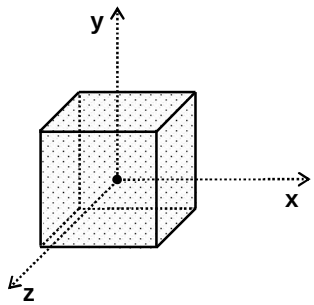
(B) The position vector of the particle at time t is given by $\vec{r} = \left[12\cos^2\left(\frac{\pi t}{8}\right)\hat{i} + 5\sin^2\left(\frac{\pi t}{8}\right)\hat{j}\right]$ m.

(C) The velocity vector of the particle at time t is given by $\vec{v} = \left[\frac{3\pi}{2}\sin\left(\frac{\pi t}{4}\right)\hat{i} - \frac{5\pi}{8}\sin\left(\frac{\pi t}{4}\right)\hat{j}\right]$ m/s.

(D) The amplitude of simple harmonic motion executed by the particle is $\frac{13}{2}$ m.

2. Choose the incorrect option(s)
   (A) The buoyant force on a submerged rigid body can be considered to be acting at the centre of mass of the rigid body.
   (B) Steady fluid flow in a tube of flow of variable cross-section is non-accelerated.
   (C) Viscosity of a liquid increases with rise in temperature.
   (D) The angle of contact of liquid in a capillary tube and surface tension of liquid decrease with increase in temperature.

3. A rubber ball is kept at rest at the centre of a cube filled with liquid. The density of liquid is ρ and rubber ball is $\rho/3$. At $t = 0$, the cube is accelerated with an acceleration $\vec{a} = \left(\frac{5}{\sqrt{2}}\hat{i} - \frac{15}{2}\hat{j} + \frac{5}{2}\hat{k}\right)$ m/s^2 and the ball is released from origin as shown in the figure. The acceleration due to gravity is along negative y-axis. At $t = 0$, choose the correct option(s). (Take g = 10 m/s^2)



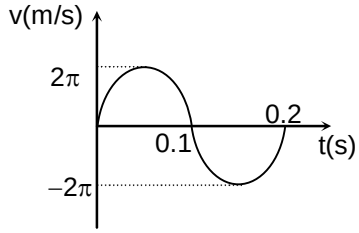
(A) Angle made by acceleration of ball with x-axis is $\frac{\pi}{4}$ as seen from the cube.

(B) Angle made by acceleration of ball with y-axis is $\frac{\pi}{3}$ as seen from the cube.

(C) Angle made by acceleration of ball with z-axis is $\frac{\pi}{3}$ as seen from the cube.

(D) If density of rubber ball is considered as $\rho/2$ then angle made by acceleration vector of ball with x, y and z-axis is changed as seen from the cube.

4. A spring block system performs simple harmonic motion about its equilibrium position along x-axis. The spring constant of used spring is 200 N/m and equilibrium position of block is origin. The velocity time graph of the particle is given in the figure. Choose the correct option(s). (Take $\pi^2 = 10$)

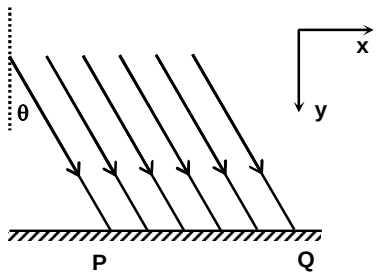


(A) The mass of the block is 0.2 kg.
(B) The displacement of the particle from equilibrium position at t = 0 is 0.1 meter.
(C) The acceleration of block at t = 0.1 second is -200 m/s^2.
(D) The maximum kinetic energy of the block is 4.0 Joule.

5. An imaginary particle has a charge equal to the charge of electron but mass of imaginary particle is 50 times the mass of electron. Now, consider that imaginary particle moves in circular orbit around a nucleus. The charge of nucleus is +4e, and nucleus is assumed to be at rest. Assume mass of nucleus is very large as compared to the mass of imaginary particle moving around it. Choose the correct option(s).

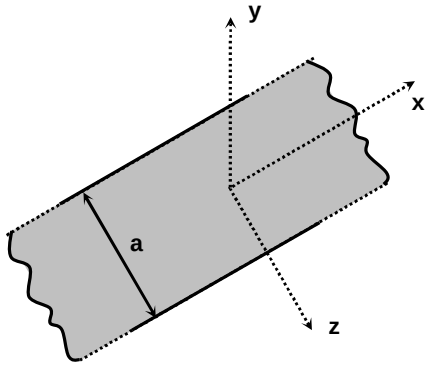
(A) The radius of nth orbit is $\frac{n^2h^2\varepsilon_0}{200\pi me^2}$, where m is mass of electron, h is Planck constant, e is charge of electron.
(B) If all atoms are at 16th excited state then during de-excitation till 10th excited state the total maximum number of possible wavelengths emitted is 21.
(C) The wavelength of photon emitted when transition takes place from third excited state to ground state is 1.21 Å.
(D) Since mass of imaginary particle is different from the mass of electron so the velocity of imaginary particle and the electron in nth orbit will be different.

6. A reflecting surface PQ is fixed and a parallel beam of light is incident on it. The intensity of light is 600 W/m^2 and it makes an angle 53° with the vertical line. The area of surface PQ is 2.0 m^2. It is observed that 40% incident energy is reflected by the reflecting surface PQ. Now choose the correct option(s).



(A) Due to incident beam force along the x-axis on the surface is 1.152×10^{-6} N.
(B) Due to incident beam force along the y-axis on the surface is 2.016×10^{-6} N.
(C) Due to incident energy radiation pressure on the surface is 0.576×10^{-6} N/m^2.
(D) Due to incident energy radiation pressure on the surface is 1.008×10^{-6} N/m^2.

7. A thin metallic plate of uniform mass density is kept along x-axis symmetrically. The length of plate along x-axis is very large as compared to its width 'a' along z-axis. It is given that 'σ' is mass per unit area. Then choose the correct option(s)



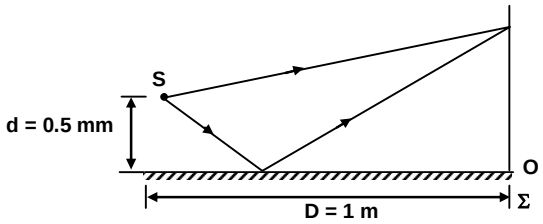
(A) The gravitational field at $\left(0,\frac{\sqrt{3}\,a}{2},0\right)$ is $\frac{2G\sigma\pi}{3}$.

(B) The gravitational field at $\left(0,\frac{a}{2},0\right)$ is $G\sigma\pi$.

(C) The ratio of gravitational field at $\left(0,\frac{a}{2\sqrt{3}},0\right)$ and $\left(0,\frac{\sqrt{3}\,a}{2},0\right)$ is 2.

(D) The ratio of gravitational field at $\left(0,\frac{a}{2\sqrt{3}},0\right)$ and $\left(0,\frac{a}{2},0\right)$ is $\frac{4}{3}$.

8. A narrow slit S transmitting light of wavelength λ is placed a distance d above a large plane mirror as shown in the figure. The light coming directly form the slit and that coming after reflection from the plane mirror interfere on a screen Σ placed at a distance D from the slit. Assume that the mirror is 100% reflecting. (Given D = 1m, d = 0.5 mm and λ = 500 nm)



(A) Distance from O, where first interference maximum is formed on the screen is $\frac{1}{4}$mm.

(B) Distance from O, where closest interference minimum is formed on the screen is zero.

(C) Fringe width of the interference pattern formed on the screen is $\frac{1}{2}$mm.

(D) Closest point to O on the screen where waves interfere with a phase difference of $\frac{\pi}{2}$ is $\frac{1}{8}$mm.

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

Two point sound sources of same frequency 'f' amplitude 2A and 4A are situated at x = 0 and x = 20 λ respectively (λ = wavelength of wave) on the x-axis. These sources are vibrating in same phase. Four observers A, B, C and D are situated at $x_A = 4\lambda$, $x_B = 8\lambda$, $x_C = 12\lambda$ and $x_D = 16\lambda$ respectively. The value of intensity observed by another observer O situated exactly at midpoint of location of two sources is I_0. Observers A, B, C and D are tabulated in first List and in second List some quantities are given. (Given $I_0 = \frac{36A^2k}{100}$)

List –I gives the above four observers while List –II lists the magnitude of some quantity.

	List-I		List-II
(I)	Observer A	(P)	$\frac{9}{16}$

(II)	Observer B	(Q)	$\frac{49}{144}$
(III)	Observer C	(R)	$\frac{4}{9}$
(IV)	Observer D	(S)	$\frac{5}{16}$
		(T)	$\frac{81}{64}$
		(U)	$\frac{10}{36}$

9. List –I contains observers and List-II contains the values of intensity observed by respective observers in unit of kA^2. Pick the correct option.

Options

(A) $I \rightarrow Q, II \rightarrow R, III \rightarrow S, IV \rightarrow P$
(B) $I \rightarrow P, II \rightarrow Q, III \rightarrow R, IV \rightarrow T$
(C) $I \rightarrow P, II \rightarrow Q, III \rightarrow T, IV \rightarrow S$
(D) $I \rightarrow Q, II \rightarrow P, III \rightarrow R, IV \rightarrow T$

10. In separate experiment location of sources are interchanged. The values of intensity observed by the observers now in units of kA^2 are given in List-II. Match respective intensities observed by the observers.

Options

(A) $I \rightarrow T, II \rightarrow R, III \rightarrow S, IV \rightarrow Q$
(B) $I \rightarrow Q, II \rightarrow S, III \rightarrow R, IV \rightarrow P$
(C) $I \rightarrow P, II \rightarrow Q, III \rightarrow T, IV \rightarrow S$
(D) $I \rightarrow T, II \rightarrow R, III \rightarrow Q, IV \rightarrow P$

Answer the following by appropriately matching the list based on the information given in the paragraph.

A uniform candle of length 10 cm is glued with an aluminium cylinder of same radius and length 1 cm. Composite system is submerged in the water and it floats stably in vertical orientation. Density of wax is 0.8 g/cm^3, density of aluminium is 2.7 g/cm^3 and density of water is 1.0 g/cm^3. At t = 0 candle is lit and assume that it consumes wax at a constant rate of 0.1 cm/min. List –I gives four different instant of time while List-II lists magnitude of some quantity.

	List-I		**List-II**
(I)	t = 5 minutes	(P)	$\sqrt{\frac{103}{107}}$
(II)	t = 10 minutes	(Q)	9.3
(III)	t = 15 minutes	(R)	8.9
(IV)	t = 20 minutes	(S)	$\sqrt{\frac{99}{107}}$
		(T)	8.5
		(U)	$\sqrt{\frac{95}{107}}$

11. The correct match for time and length of the candle inside the water (in cm) will be:

Options

(A) $I \rightarrow Q, II \rightarrow R, III \rightarrow T, IV \rightarrow T$

(B) I → Q, II → S, III → P, IV → R
(C) I → P, II → S, III → Q, IV → T
(D) I → P, II → P, III → S, IV → T

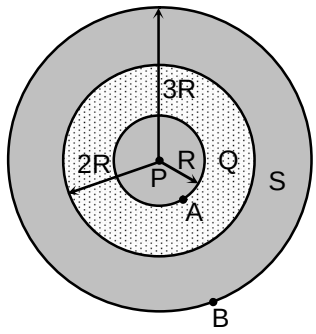
12. If candle is extinguished at different times i.e. at t = 5 minute, t = 10 minute, t = 15 minute and at t = 20 minutes and slightly displaced from mean position in each case. The correct match for time period of small oscillation in each case will be (in units of T_0 where T_0 is time period at t = 0)
Options
(A) I → P, II → P, III → U, IV → U
(B) I → Q, II → U, III → T, IV → T
(C) I → Q, II → P, III → S, IV → S
(D) I → P, II → S, III → U, IV → U

SECTION – C
(Numerical Answer Type)

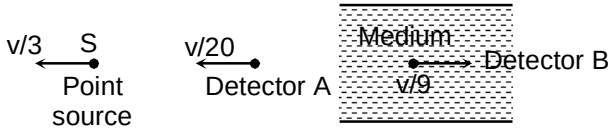
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. The sound from a trumpet is emitted uniformly in all directions at 20°C. At point 'P' in space the sound intensity level is 30 dB. The density of air in the medium is 1.5 kg/m^3, the velocity of sound at 20°C is 300 m/sec. Find the displacement amplitude (in Å) at point 'P'. The frequency of sound wave in the medium is 600 Hz. (Take $\pi^2 = 10$)

14. A solid sphere P and two hollow sphere Q and S are concentrically placed as shown in the figure. The radius of P. Q and S are R, 2R and 3R respectively. The density of P, Q and S are ρ_1, ρ_2 and ρ_3 respectively. It is given that the density of middle sphere is the arithmetic mean of the densities of other two spheres. If the gravitational fields at points A and B are equal then find the ratio of $\frac{\rho_3}{\rho_1}$.

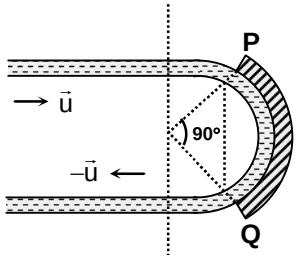


15. A point source 'S' is moving in air towards left with a constant velocity $\frac{v}{3}$, where v is the speed of sound in air. There are two detectors A and B. A is moving in air with a velocity $\frac{v}{20}$ towards left and B is moving in a medium with a velocity $\frac{v}{9}$ towards right as shown in the figure. The velocity of sound in the medium is 4V and frequency of sound emitted by the point source is f. Let λ_A is the wavelength detected by the detector A and λ_B the wavelength detected by the detector B. If $\lambda_B - \lambda_A = k\left(\frac{v}{3f}\right)$, then find the value of k.

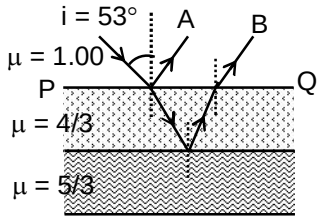


16. A small cube of negligible mass is floating on the surface of lake. The side of the cube is 30 cm and depth of lake is 2m. The cube is pushed very slowly under the water, down to the bottom of the lake. The density of water is 10^3 kg/m^3 and acceleration due to gravity is 9.8 m/s^2. Find the work done (in Joule) by external pushing force in the process.

17. PQ is a stationary circular curved surface blade. A stream line water impinges on the curved surface of the blade. The speed of stream line water is $5\sqrt{2}$ m/s before and after it strikes the curved surface blade. The mass flow rate of water per unit time is 112 gm/sec. Find the force (in newton) exerted by the stream line water on the blade.



18. A light wave of wavelength λ = 650 nm is incident on the surface PQ at an angle of incidence i = 53° as shown. If the phase difference between the two reflected rays A and B is π. Find the thickness (in nanometer) of the thin film of refractive index $\frac{4}{3}$.



Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. The correct statement(s) among the following is/are
    (A) The complexes $\left[Co(NH_3)_6\right]^{3+}$ and $\left[Fe(NH_3)_6\right]^{2+}$ have same geometry
    (B) The complexes $\left[Co(NH_3)_6\right]^{3+}$ and $\left[Fe(NH_3)_6\right]^{2+}$ differ in their magnetic properties
    (C) The complexes $\left[Pt(CN)_4\right]^{2-}$ and $\left[CuCl_4\right]^{2-}$ differ in their geometry
    (D) The complexes $\left[Ni(NH_3)_6\right]^{2+}$ and $\left[Cr(NH_3)_6\right]^{3+}$ differ in their primary valency

20. The correct statement(s) among the following is/are
    (A) In 3d-series enthalpy of atomization of Zn is lowest.
    (B) $Cu^{2+}(aq)$ is more stable than $Cu^{+}(aq)$
    (C) Sc does not exhibit variable oxidation state
    (D) MnO_4^- is coloured due to d – d transition of electron.

21. Which of the following order given below is/are incorrect for the property indicated against it?
    (A) $OCl^- < ClO_2^- < ClO_3^- < ClO_4^-$ (Cl – O bond distance)
    (B) $HClO_4 < HBrO_4 < HIO_4$ (Acidity)
    (C) Cl > F > Br > I (Electron gain enthalpy)
    (D) H – F < H – Cl < H – Br < H – I (Bond energy)

22. The correct statement(s) among the following is/are
    (A) Cu^{2+} (aq) ions react with $K_4[Fe(CN)_6]$ to form chocolate brown precipitate.
    (B) Cu^{2+} (aq) ions react with $K_3[Fe(CN)_6]$ to form green precipitate.
    (C) Hg_2Cl_2 when treated with NH_4OH gives a black residue.
    (D) NH_4Cl solution when treated with Nessler's reagent gives green precipitate.

23. The incorrect statement(s) among the following is/are
    (A) Red phosphorous is more reactive than white phosphorous
    (B) P_2H_4 is more stable than N_2H_4
    (C) Bond angle in PH_3 is higher than that in PH_4^+
    (D) In the solid state PBr_5 exists as $\left[PBr_4\right]^+\left[PBr_6\right]^-$

24. Which of the following ore(s) is/are concentrated by froth floatation process
    (A) Galena
    (B) Copper pyrite
    (C) Zinc blende
    (D) Bauxite

25. Which of the following molecule(s) have cyclic structure?
    (A) $N_3P_3Cl_6$
    (B) C_3N_{12}
    (C) N_4S_4
    (D) $P_3N_2Cl_9$

26. Which of the following statement(s) is/are correct about azeotropic mixture?
(A) Azeotropic mixture are non ideal solution
(B) Solution with large negative deviation from Raoult's law forms maximum boiling azeotropes
(C) The components of azeotropic mixture cannot be separated completely by fractional distillation
(D) Azeotropes obey Raoult's law

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

List-I includes the complex ions and List-II includes the hybridization of central atom in the complex ions and magnetic property of the complex ions:

LIST–I		LIST–II	
(I)	$[CoF_6]^{3-}$	(P)	sp^3
(II)	$[Fe(CN)_6]^{4-}$	(Q)	dsp^2
(III)	$[Co(NH_3)_6]^{3+}$	(R)	sp^3d^2 (outer orbital complex)
(IV)	$[CuCl_4]^{2-}$	(S)	d^2sp^3 (inner orbital complex)
		(T)	Paramagnetic
		(U)	Diamagnetic

27. Which of the following options has the correct combination considering List-I and List-II:
(A) (I), (U), (S)
(B) (III), (S), (U)
(C) (II), (T), (S)
(D) (IV), (Q), (T)

List-I includes the complex ions and List-II includes the hybridization of central atom in the complex ions and magnetic property of the complex ions:

LIST–I		LIST–II	
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(III)	$[Co(NH_3)_6]^{3+}$	(R)	sp^3d^2 (outer orbital complex)
(IV)	$[CuCl_4]^{2-}$	(S)	d^2sp^3 (inner orbital complex)
		(T)	Paramagnetic
		(U)	Diamagnetic

28. Which of the following options has the correct combination considering List-I and List-II:
(A) (I), (R), (T)
(B) (III), (Q), (T)
(C) (II), (R), (T)
(D) (IV), (P), (U)

Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I includes the oxides of nitrogen and List-II includes the properties of the oxides of nitrogen.

LIST–I		LIST–II	
(I)	N_2O_5	(P)	Colourless gas
(II)	N_2O	(Q)	Brown coloured gas
(III)	NO	(R)	Acidic oxide
(IV)	NO_2	(S)	Neutral oxide
		(T)	Blue coloured solid
		(U)	Paramagnetic

29. Which of the following options has the correct combination considering List-I and List-II
(A) (IV), (R), (U), (Q)
(B) (III), (T), (S), (U)
(C) (II), (Q), (U), (S)
(D) (I), (P), (R), (U)

Answer the following by appropriately matching the list based on the information given in the paragraph.

List-I includes the oxides of nitrogen and List-II includes the properties of the oxides of nitrogen.

List–I		List–II	
(I)	N_2O_5	(P)	Colourless gas
(II)	N_2O	(Q)	Brown coloured gas
(III)	NO	(R)	Acidic oxide
(IV)	NO_2	(S)	Neutral oxide
		(T)	Blue coloured solid
		(U)	Paramagnetic

30 Which of the following options has the correct combination considering List-I and List-II
(A) (II), (T), (U), (S)
(B) (I), (Q), (U), (S)
(C) (III), (P), (S), (U)
(D) (IV), (P), (U), (R)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The rate constant for the forward reaction
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 92.12×10^{-2} min^{-1}. The time (in sec) required for the molar ratio of NO_2 (g) to N_2O_4 (g) in the reaction mixture to become 2 : 1 is [log 2 = 0.3010]

32. $NH_3(g)$ reacts with CuO (s) according to the following reaction

$$3CuO(s) + 2NH_3(g) \longrightarrow 3Cu(s) + N_2(g) + 3H_2O(\ell)$$

Standard enthalpies of formation of $NH_3(g), CuO(s)$ and $H_2O(\ell)$ are -46, - 155 and – 285 kJ/mole respectively. The heat liberated (in kJ) when 2.72 g of NH_3(g) is reacted completely with CuO(s) according to the reaction given above when all the reactants and products are in their standard state is

33. At higher altitude, boiling point of water is 95°C. The mass in gram of NaCl $(\alpha = 0.8)$ added to 100 g of water (K_b = 0.52 K kg mol^{-1}) in order to raise the boiling point to 100°C is (MW of NaCl = 58.5)

34. 3 g of an oxide of a metal is converted completely to 5 g chloride of the same metal. Equivalent weight of metal is

35. The density of a certain gaseous oxide at 3.2 bar pressure and 47°C is same as that of O_2 at 27°C and 4.2 bar pressure. The molar mass of the gaseous oxide is

36. The specific conductivity of a saturated solution of AgCl is 3.38×10^{-6} ohm^{-1} cm^{-1} at 25°C. If $\lambda^o_{Ag^+} = 62.3$ $ohm^{-1} cm^2 mol^{-1}$ and $\lambda^o_{Cl^-} = 67.7$ $ohm^{-1} cm^2$ mol^{-1}, then the solubility of AgCl at 25°C is $y \times 10^{-5}$ M. The value of y is

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If the equation $x^4 + px^3 + qx^2 + rx + 5 = 0$ has four positive real roots, then
(A) p can be –10
(B) r can be –20
(C) $pr \geq 80$
(D) none of these

38. If $a, b \in R$ and $a + b \geq 0$, let $P = (a^2 + b^2)^3$ and $Q = 32(a^3 + b^3)(ab - a - b)$, then
(A) $P < Q \ \forall$ a, b
(B) $P \leq Q \ \forall$ a, b
(C) $P \geq Q \ \forall$ a, b
(D) none of the above

39. If $y = \sum_{r=1}^{n} (-1)^{r-1} \left(\frac{r^2 + 3r + 1}{r^3 + 2r^2 + r} \right) {}^nC_r$, then which of the following is correct?
(A) $y < \frac{n+1}{n}$
(B) $y = \frac{n+1}{n}$
(C) $y \geq \sum_{r=1}^{n} \left(\frac{n+1}{n} \right) \left(\frac{1}{r} \right)$
(D) $y \leq \sum_{r=1}^{n} \left(\frac{n}{n+1} \right) \left(\frac{1}{r} \right)$

40. If $|z+2| - |z-2| = 2$ and $\frac{z-a}{z+a} = ik$, where k is a real parameter, $a \in R$ has exactly two solutions, then
(A) $a < -1$
(B) $a > 1$
(C) $a = 1$ only
(D) $a = -1$ only

41. A line L with direction cosines l, m, n meets each of the lines $y = x \tan\alpha$, $z = c$; $y = -x \tan\alpha$, $z + c = 0$, then the shortest distance between the line L and the z-axis is
(A) $\left| \frac{c \tan\alpha (m^2 - l^2)}{n\sqrt{m^2 + l^2}} \right|$
(B) $\left| \frac{c (m^2 \cot\alpha - l^2 \tan\alpha)}{n\sqrt{m^2 + l^2}} \right|$
(C) $\left| \frac{c (m^2 \tan^2\alpha - l^2 \cot^2\alpha)}{n\sqrt{m^2 + l^2}} \right|$

(D) $\left|\frac{c(m^2-l^2)}{n\sqrt{m^2+l^2}}\right|$ for $\alpha = \frac{\pi}{4}$

42. Let the line L on the plane x + y + z = 1, such that a plane perpendicular to x + y + z = 1 contains the line L and $\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-1}{1}$, then
(A) line L passes through the point (1, –1, 1)
(B) line L passes through the point $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$
(C) one of the direction ratio of line L is (1, –1, –1)
(D) one of the direction ratio of line L is (2, –1, –1)

43. Let a, b, c, d can be arranged at 'n' places such that a occurs exactly even number of times, then the number of such arrangements are
(A) 28 for n = 3
(B) $2^{n-1}(2^n - 1)$ for $n \in N$
(C) 36 for n = 3
(D) $2^{n-1}(2^n + 1)$ for $n \in N$

44. If the determinant $A = \begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix} = 0$, then
(A) a, b, c are in Arithmetic Progression
(B) a, b, c are in Geometric Progression
(C) α is root of equation $ax^2 + bx + c = 0$
(D) α is root of equation $ax^2 + 2bx + c = 0$

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

MM-I. Let $x_i = \frac{a_i}{30} \forall$ i = 1, 2, 3, ..., n is a rational number in its lowest form and $S_{(k,r)} = \sum_{i=1}^{n} x_i$, $k < x_i < r$

LIST–I		LIST–II	
(I)	$S_{(0,1)}$ is equal to	(P)	2
(II)	$S_{(0,2)}$ is equal to	(Q)	4
(III)	$S_{(n,2n)}$ is equal to	(R)	14
(IV)	$S_{(0,n)}$ is equal to	(S)	$2n^2$
		(T)	$4n^2$
		(U)	$14n^2$

45. Which of the following is the only CORRECT combination?
(A) (I), (P)
(B) (I), (Q)
(C) (II), (Q)
(D) (II), (R)

46. Which of the following is the only CORRECT combination?
(A) (III), (S)
(B) (III), (U)
(C) (IV), (T)
(D) (IV), (U)

Answer the following by appropriately matching the list based on the information given in the paragraph.

MM-II. Let position vectors of the point A, B, C are $\bar{a} = \hat{i} + \hat{j} + \hat{k}$, $\bar{b} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\bar{c} = 5\hat{i} + \hat{j} - \hat{k}$ respectively, and let position vector of a variable point R be $\bar{r}$

LIST–I		LIST–II	
(I)	locus of point $R(\bar{r})$ such that $\bar{r} \cdot \bar{a} = 5$	(P)	pair of lines
(II)	locus of point $R(\bar{r})$ such that $\lvert\bar{r} - \bar{b}\rvert + \lvert\bar{r} - \bar{c}\rvert = 4$	(Q)	circle
(III)	locus of point $R(\bar{r})$ such that $\bar{r} \cdot \bar{a} = 5$ and $\lvert\bar{r} - \bar{b}\rvert + \lvert\bar{r} - \bar{c}\rvert = 4$	(R)	ellipse
(IV)	area of the region formed by point $R(\bar{r})$ such that $\bar{r} \cdot \bar{a} = 5$ and $\lvert\bar{r} - \bar{b}\rvert + \lvert\bar{r} - \bar{c}\rvert = 4$	(S)	π
		(T)	$\sqrt{2}\,\pi$
		(U)	2π

47. Which of the following is the only CORRECT combination?
(A) (I), (Q)
(B) (I), (R)
(C) (III), (R)
(D) (III), (Q)

48. Which of the following is the only CORRECT combination?
(A) (II), (Q)
(B) (II), (P)
(C) (IV), (U)
(D) (IV), (T)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $n = 2^{31}.3^{19}$, then number of factors of n^2 less than n and does not divide n are

50. A person A was issued a 10 digit mobile phone number having his 6 digit birth date (DDMMYY format) in it. Let E be the event that his birth was in first 9 days of a month, then the probability of the occurrence of the event E is

51. Let $y = \sum_{r=1}^{\infty} \frac{1.3.5.....(2r-1)}{r!}\left(\frac{2}{5}\right)^r$, then $\frac{(y+1)^2}{2}$ is

52. Number of ordered pair(s) (x, y) that satisfies the inequation $y - |x| \geq 1+\sqrt{x^2+y^2-1}$ is/are

53. If $a(n) = x^n + y^n$, where $x, y \in [-1, 1]$ and $x^2 + y^2 = 1$, then maximum value of $\frac{a(n)-a(n+2)}{a(n-2)}$ is

54. If a plane passes through the intersection of $5x + b_1y + c_1z = d_1$ and $4x + b_2y + c_2z = d_2$ and contains the x-axis, then $d_1 : d_2$ is